

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change]	262935	74.161876
1	[Java, ByteCode, Member, Method]	13527	3.815345
2	[Java, ByteCode, Parameter]	13296	3.750190
3	[Git, Commit]	12955	3.654010
4	[File, Git]	10908	3.076645
5	[Java, ByteCode, Bound, ParameterizedType]	7279	2.053071
6	[Java, ByteCode, Bound]	7235	2.040661
7	[Java, ByteCode, Member, Field]	3602	1.015959
8	[Java, ByteCode, Bound, WildcardType]	2947	0.831213
9	[Java, Value, ByteCode, Annotation]	2931	0.826700
10	[Xml, Element]	2162	0.609801
11	[Java, ByteCode, Member, Constructor, Method]	2129	0.600493
12	[Xml, Text]	1450	0.408978
13	[Java, ByteCode, TypeVariable, Bound]	1111	0.313362
14	[Java, ByteCode, Member, Method, Lambda]	972	0.274157
15	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.250464
16	[Type, File, Java, Class, ByteCode]	846	0.238618
17	[Json, Key]	702	0.198002
18	[Value, Json, Scalar]	685	0.193207
19	[Java, Value, ByteCode, Primitive]	672	0.189540
20	[Type, File, Java, ByteCode, JavaType]	649	0.183053
21	[Java, ByteCode, Member, Method, GenericDeclar...	578	0.163027
22	[Type, File, Java, ByteCode, ExternalType]	411	0.115924
23	[Author, Git, Person]	299	0.084334
24	[Value, Array]	287	0.080950
25	[Committer, Git, Person]	252	0.071078
26	[Type, File, Java, Class, ByteCode, GenericDec...	237	0.066847
27	[Value, Property]	211	0.059513
28	[Type, File, Java, ByteCode, Interface]	190	0.053590
29	[Java, Value, Class, ByteCode]	189	0.053308

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

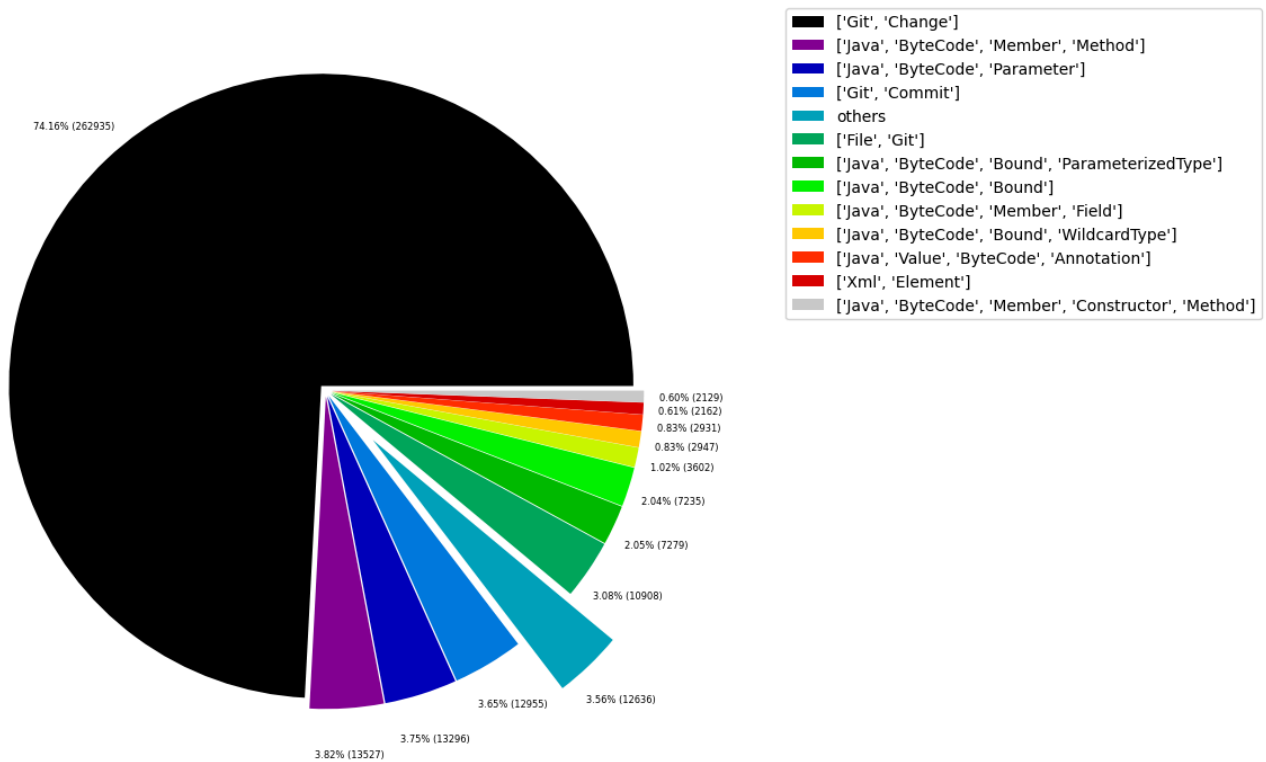


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000282
1	[Package, File, Json, NPM]	1	0.000282
2	[Repository, File, Git]	1	0.000282
3	[File, TS, Scan]	1	0.000282
4	[File, Json]	2	0.000564
5	[File]	3	0.000846
6	[Java, ByteCode, Member, Constructor, Method, ...]	4	0.001128
7	[Maven, Exclusion]	5	0.001410
8	[Value, Array, Json]	6	0.001692
9	[Dependency, NPM]	7	0.001974
10	[Type, File, Java, ByteCode, Void]	9	0.002538
11	[File, Maven, Xml, Pom, Document]	9	0.002538
12	[Java, ManifestSection]	9	0.002538
13	[File, Java, Manifest]	9	0.002538
14	[Artifact, File, Jar, Archive, Zip, Java]	9	0.002538
15	[File, Java, ServiceLoader]	10	0.002821
16	[File, Java, Properties]	12	0.003385
17	[Maven, PluginExecution]	16	0.004513
18	[Maven, ExecutionGoal]	16	0.004513
19	[Xml, Attribute]	18	0.005077
20	[jQAssistant, Rule, Concept]	19	0.005359
21	[Maven, Configuration]	21	0.005923
22	[Maven, Plugin]	21	0.005923
23	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.005923
24	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.006205
25	[Type, File, Java, ByteCode, Enum]	28	0.007898
26	[Type, File, Java, ByteCode, PrimitiveType]	30	0.008462
27	[Xml, Namespace]	36	0.010154
28	[Type, File, Java, ByteCode, Annotation]	44	0.012410
29	[Git, Branch]	46	0.012974

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

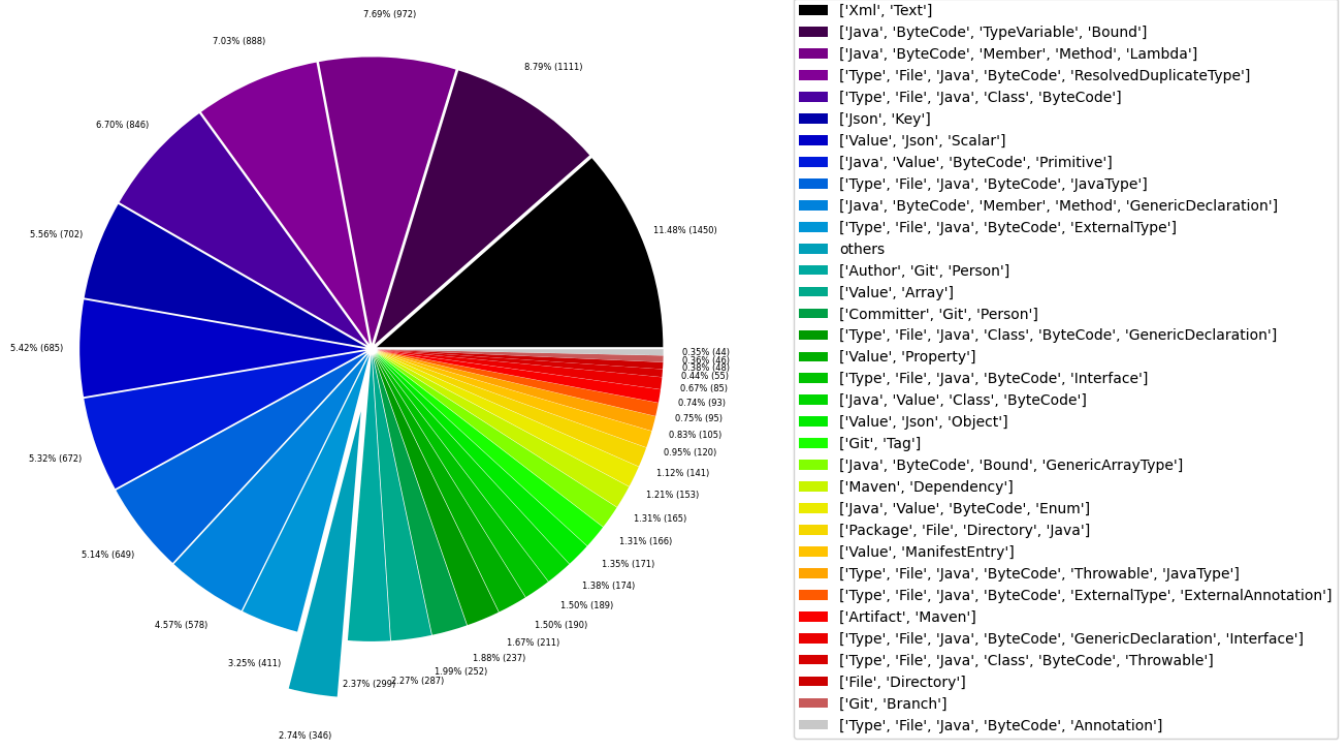


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

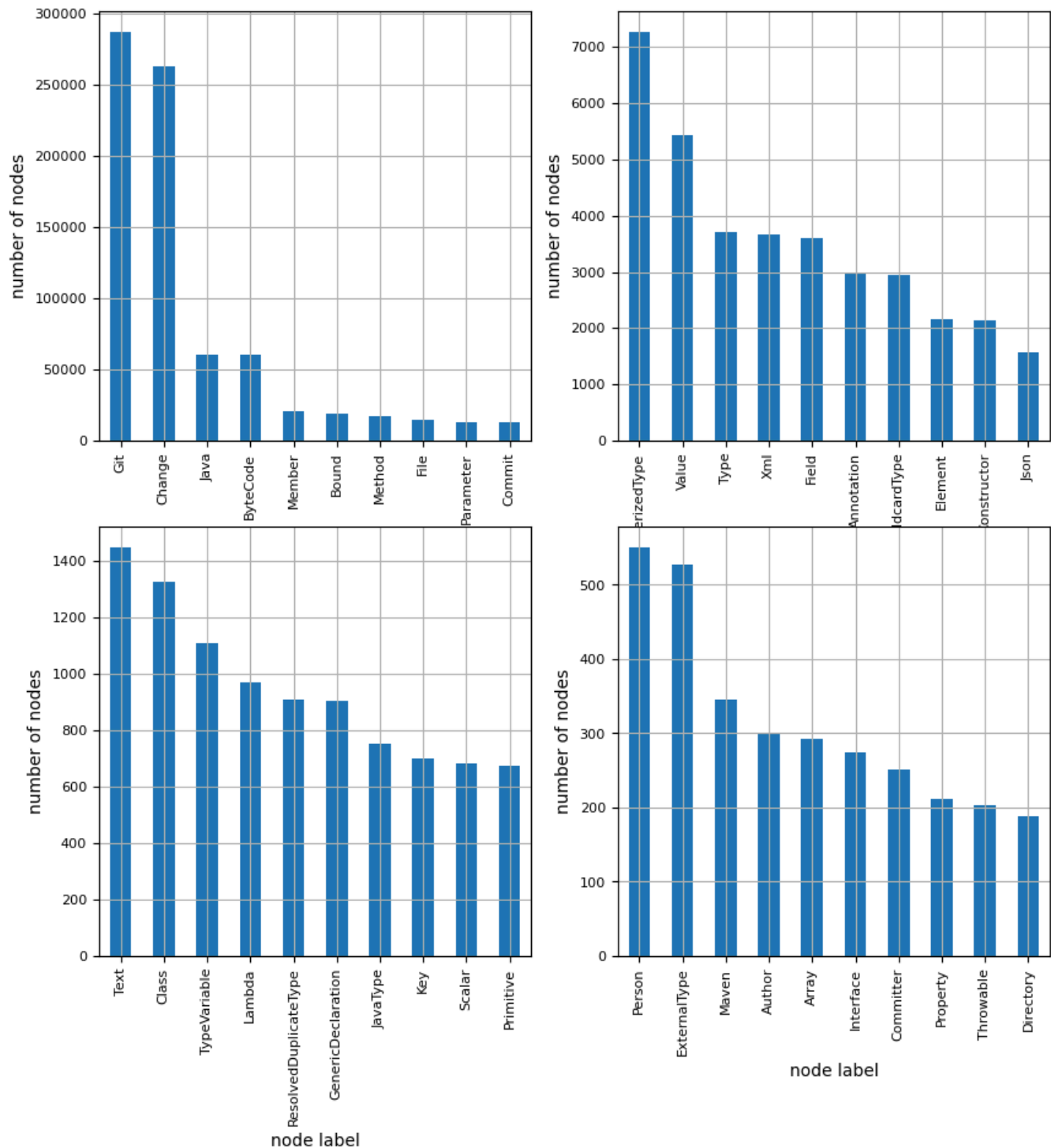
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	287567	81.109431
1	Change	262935	74.161876
2	Java	60696	17.119551
3	ByteCode	60506	17.065961
4	Member	20812	5.870108
5	Bound	18738	5.285128
6	Method	17210	4.854150
7	File	14869	4.193861
8	Parameter	13296	3.750190
9	Commit	12955	3.654010
10	ParameterizedType	7279	2.053071
11	Value	5428	1.530989
12	Type	3715	1.047831
13	Xml	3675	1.036549
14	Field	3602	1.015959
15	Annotation	2975	0.839111
16	WildcardType	2947	0.831213
17	Element	2162	0.609801
18	Constructor	2133	0.601621
19	Json	1570	0.442825
20	Text	1450	0.408978
21	Class	1327	0.374286
22	TypeVariable	1111	0.313362
23	Lambda	972	0.274157
24	ResolvedDuplicateType	910	0.256669
25	GenericDeclaration	904	0.254977
26	JavaType	754	0.212669
27	Key	702	0.198002
28	Scalar	685	0.193207
29	Primitive	672	0.189540
30	Person	551	0.155412
31	ExternalType	527	0.148642
32	Maven	346	0.097591
33	Author	299	0.084334
34	Array	293	0.082642
35	Interface	275	0.077565
36	Committer	252	0.071078
37	Property	211	0.059513
38	Throwable	203	0.057257
39	Directory	189	0.053308

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1100176

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	262935	23.899358
1	MODIFIES	262935	23.899358
2	UPDATES	173508	15.770931
3	CREATES	61941	5.630099
4	DELETES	38052	3.458719
5	INVOKES	36628	3.329285
6	COMMITTED	25910	2.355078
7	DEPENDS_ON	22488	2.044037
8	OF_TYPE	21922	1.992590
9	DECLARES	21273	1.933600
10	OF_RAW_TYPE	17348	1.576839
11	HAS_PARENT	15687	1.425863
12	HAS	14432	1.311790
13	HAS_COMMIT	12955	1.177539
14	RETURNS	12853	1.168268
15	HAS_FILE	10908	0.991478
16	RENAMES	10566	0.960392
17	READS	9378	0.852409
18	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.764150
19	HAS_NEW_NAME	6242	0.567364
20	OF_GENERIC_TYPE	5991	0.544549
21	RESOLVES_TO	5277	0.479651
22	SIMILAR	4078	0.370668
23	WRITES	3944	0.358488
24	CONTAINS	3936	0.357761
25	RETURNS_GENERIC	3593	0.326584
26	ANNOTATED_BY	2919	0.265321
27	REQUIRES	2230	0.202695
28	HAS_FIRST_CHILD	2162	0.196514
29	HAS_LAST_CHILD	2162	0.196514

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

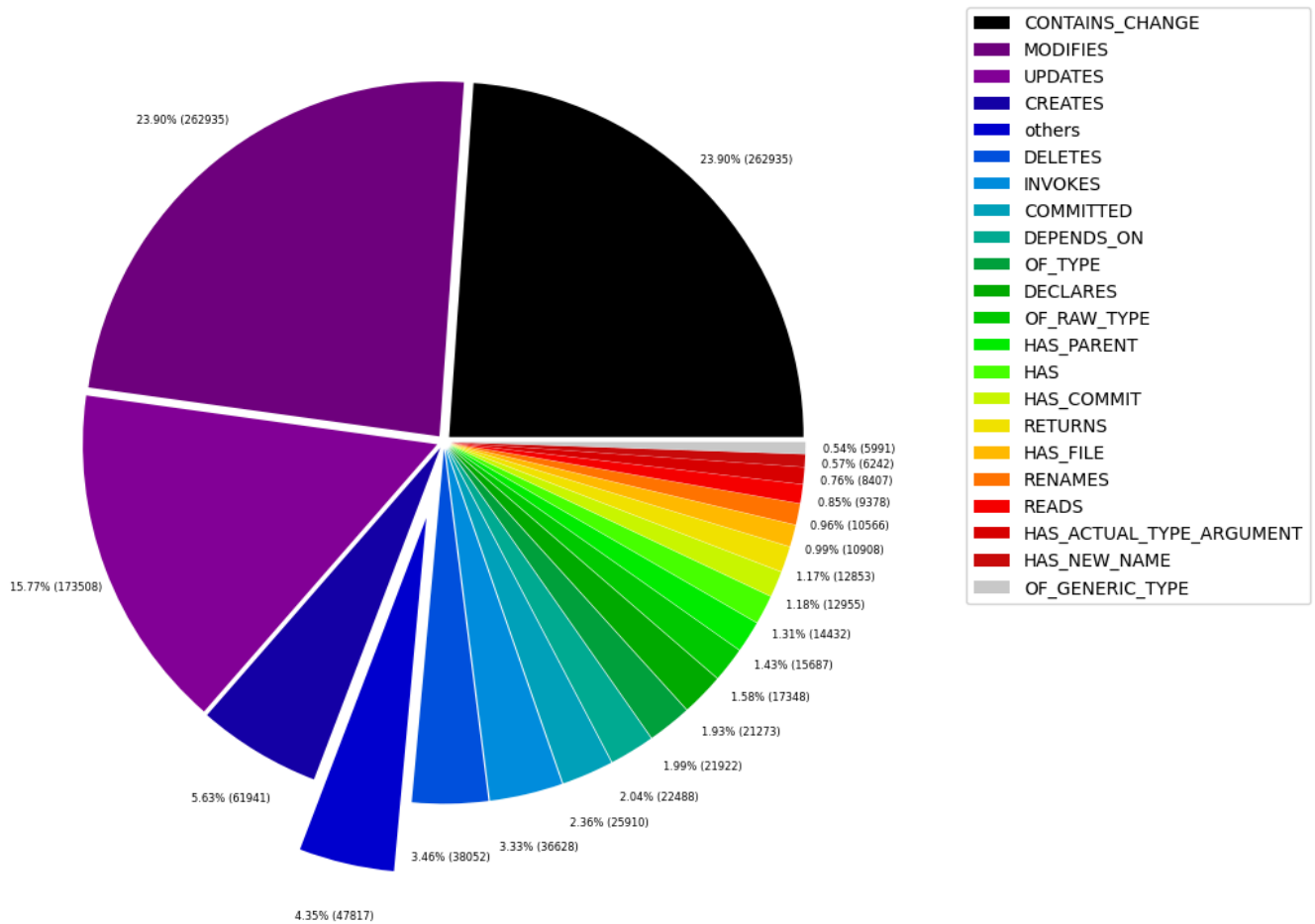


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

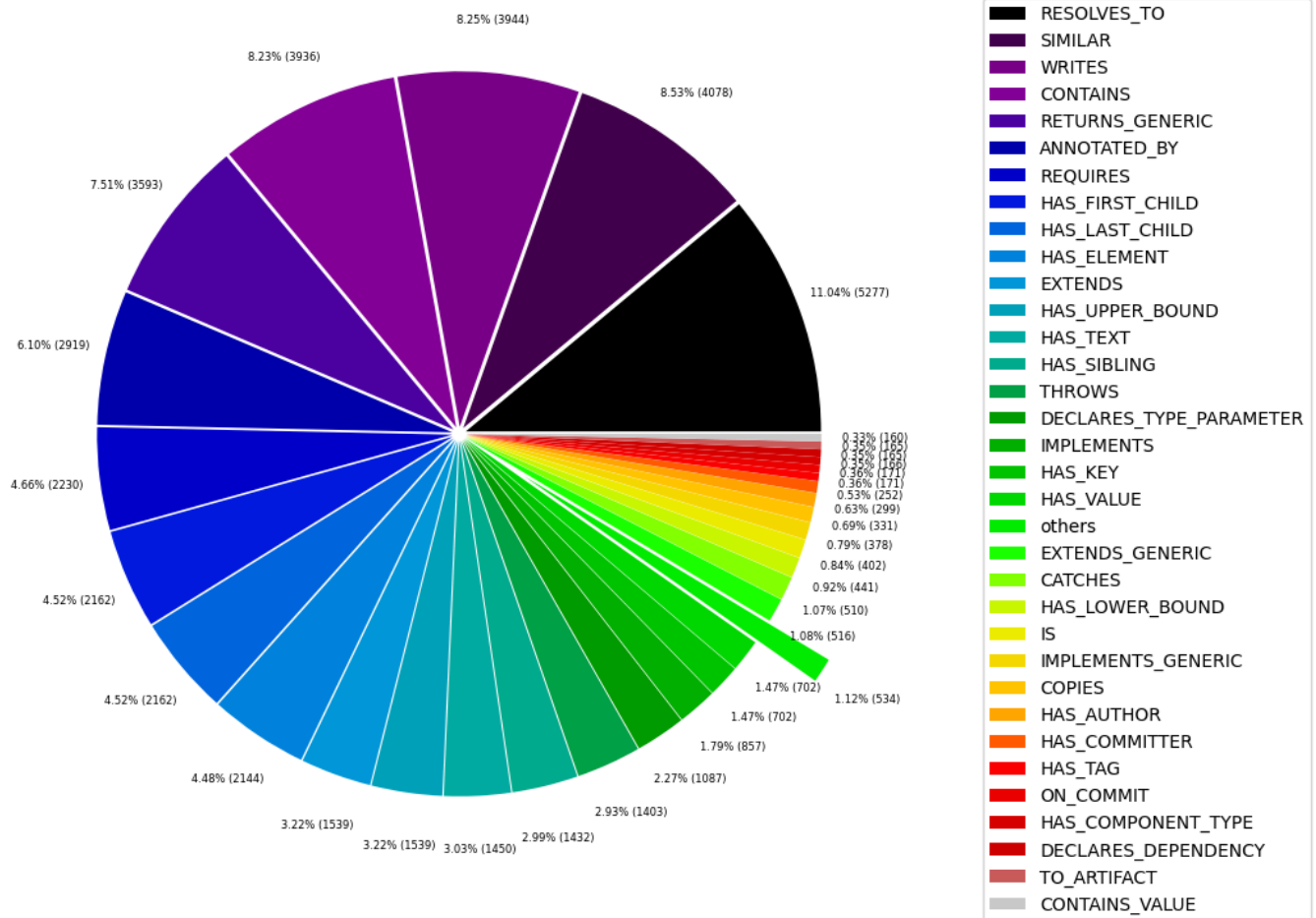
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000091
1	THROWS_GENERIC	5	0.000454
2	EXCLUDES	5	0.000454
3	DECLARES_DEV_DEPENDENCY	7	0.000636
4	DESCRIBES	9	0.000818
5	HAS_ROOT_ELEMENT	11	0.001000
6	HAS_GOAL	16	0.001454
7	HAS_EXECUTION	16	0.001454
8	OF_NAMESPACE	18	0.001636
9	HAS_ATTRIBUTE	18	0.001636
10	INCLUDES_CONCEPT	19	0.001727
11	IS_ARTIFACT	21	0.001909
12	HAS_CONFIGURATION	21	0.001909
13	USES_PLUGIN	21	0.001909
14	REQUIRES_TYPE_PARAMETER	24	0.002181
15	REQUIRES_CONCEPT	28	0.002545
16	DECLARES_NAMESPACE	36	0.003272
17	HAS_DEFAULT	39	0.003545
18	HAS_BRANCH	46	0.004181
19	HAS_HEAD	47	0.004272
20	COPY_OF	126	0.011453
21	CONTAINS_VALUE	160	0.014543
22	DECLARES_DEPENDENCY	165	0.014998
23	TO_ARTIFACT	165	0.014998
24	HAS_COMPONENT_TYPE	166	0.015088
25	ON_COMMIT	171	0.015543
26	HAS_TAG	171	0.015543
27	HAS_COMMITTER	252	0.022905
28	HAS_AUTHOR	299	0.027177
29	COPIES	331	0.030086

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Change]	MODIFIES	[File, Git]	262935	262935	
1	[Git, Commit]	CONTAINS_CHANGE	[Git, Change]	262935	12955	
2	[Git, Change]	UPDATES	[File, Git]	173508	262935	
3	[Git, Change]	CREATES	[File, Git]	61941	262935	
4	[Git, Change]	DELETES	[File, Git]	38052	262935	
5	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	22255	13428	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	15678	12955	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	12955	1	
8	[Committer, Git, Person]	COMMITTED	[Git, Commit]	12955	252	
9	[Author, Git, Person]	COMMITTED	[Git, Commit]	12955	299	
10	[Repository, File, Git]	HAS_FILE	[File, Git]	10908	1	
11	[Git, Change]	RENAMES	[File, Git]	10566	262935	
12	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8495	13428	
13	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8420	13428	
14	[File, Git]	HAS_NEW_NAME	[File, Git]	6242	10908	
15	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6160	13296	
16	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465	7235	
17	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110	7279	
18	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947	7279	
19	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2695	13296	
20	[Java, ByteCode, Member, Constructor, Method]	WRITES	[Java, ByteCode, Member, Field]	2519	2129	
21	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, TypeVariable, Bound]	2474	7279	
22	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalType, Ext...	2445	2931	
23	[Java, ByteCode, Member, Constructor, Method]	INVOKES	[Java, ByteCode, Member, Constructor, Method]	2195	2129	
24	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2144	2162	
25	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Constructor, Method]	2113	13428	
26	[Java, ByteCode, Member, Method]	RETURNS	[Type, File, Java, ByteCode, JavaType]	2107	13428	
27	[Java, ByteCode, Member, Constructor, Method]	HAS	[Java, ByteCode, Parameter]	2075	2129	
28	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound]	2056	13296	
29	[Java, ByteCode, Member, Method, Lambda]	INVOKES	[Java, ByteCode, Member, Method]	1954	972	

Graph Density

```
total_number_of_nodes (vertices): 354542
total_number_of_relationships (edges): 1100176
-> total directed graph density: 8.752417204395822e-06
-> total directed graph density in percent: 0.0008752417204395822
```